

Mathematician and algorithms engineer with graduate training in combinatorial optimization, probability, and low-dimensional topology, and hands-on experience building and deploying production-grade optimization and privacy-engineering systems. Comfortable moving between rigorous mathematical formulation and scalable implementation.

EXPERIENCE

Algorithms Engineer Jan 2026 – May 2026
Oblivious Software
(Product line discontinued due to investor-driven restructuring)

- Designed and implemented core components of a production differential-privacy SQL engine, including noise mechanism calibration via sensitivity analysis to provide rigorous privacy guarantees (Rényi DP accounting).
- Built a configurable SQL query generator with tunable pruning strategies, reducing a configuration space of roughly 3.5 million candidate queries to about 11,000–12,000 representative cases for fuzz-testing the DP-SQL engine, bug discovery, and regression testing.
- Applied Python AST manipulation and static analysis (Semgrep, CodeQL) to enforce DP-safe execution environments; contributed to OpenDP framework integration.

Associate Policy Counsel Nov 2020 – May 2022
Internet Freedom Foundation, New Delhi

- Led data-driven policy analysis on India's emerging data-regulation framework, combining quantitative research with cross-disciplinary stakeholder engagement.
- Co-authored a published analysis of the NDHM Health Data Management Policy (SSRN 3947598, 2021).

Associate Analyst Oct 2019 – Feb 2020
PRS Legislative Research, New Delhi

- Conducted quantitative research and prepared analytical briefings for Members of Parliament; coordinated multi-stakeholder consultations.

PROJECTS

Combinatorial Optimization — QUBO/Max-Cut and Vehicle Routing (VRPTW)

- Implemented multiple solution approaches to Max-Cut via Quadratic Unconstrained Binary Optimization: simulated annealing with D-Wave's Neal sampler and exact MILP formulations in Gurobi, benchmarked against classical heuristics on GSET instances.
- Built a Vehicle Routing Problem with Time Windows solver using Google OR-Tools, modeling vehicle capacity, delivery time-window, and route-continuity constraints, with custom heuristics for solution initialization, validated against Solomon benchmark instances.

CNN-Based Chess Position Evaluation Engine

- Designed and trained convolutional neural network models over multiple board-state feature encodings, evaluating tradeoffs in training/validation convergence across encoding schemes.
- Built a multi-input architecture combining spatial CNN features with auxiliary vector inputs, applying batch normalization and dropout for generalization; conducted systematic hyperparameter tuning and sensitivity analysis.

MWEM Differentially Private Synthetic Data Generation

- Implemented the Multiplicative Weights Exponential Mechanism for synthetic data generation, managing numerical stability in iterative optimization and validating privacy-utility tradeoffs empirically against theoretical bounds.

EDUCATION

M.Sc. in Mathematics 2023 – 2025
Indian Institute of Technology Delhi
Coursework: Algebraic Topology, Fourier Analysis, Functional Analysis, Probability & Statistics, Numerical Analysis, Linear Optimisation, Cryptography, Data Mining, General Relativity
Master's Thesis — Low-Dimensional Topology
Advisor: Dr. Biplab Basak, IIT Delhi · Nominated for Best Thesis Award. Investigated complexity bounds on triangulations of lens spaces via linking/discriminant forms of plumbing manifolds and discrete Morse theory; verified results computationally in Regina.

B.A. in Mathematics 2015 – 2019
Cornell University, Ithaca NY
Coursework: Riemannian Geometry, Geometry of Discrete Groups, Quantum Information Processing, Differentiable Manifolds, Discrete Structures, Mathematical Foundations for the Information Age
Course Assistant, Calculus I (Spring 2017): graded homework and led weekly discussions for 40 students.

QUALIFYING EXAM

IIT JAM — Mathematics
All-India Rank 135 (2023, self-study)

SKILLS

Optimization & OR
Gurobi Google OR-Tools D-Wave / Neal QUBO

Formal Methods
Semgrep CodeQL Python AST

Privacy & Statistics
Rényi DP MWEM OpenDP

Languages
Python C++ C SQL Java

Mathematics
Algebraic Topology Functional Analysis Numerical Analysis Computational Topology (Regina)

Tools
Git/GitHub Linux AWS LaTeX